

Zoom in论文调研

Zoom-In思考

表面理解：放大图像区域，"把图像的某个区域裁剪出来、放大、重新看"

深层本质：视觉带宽的动态重分配

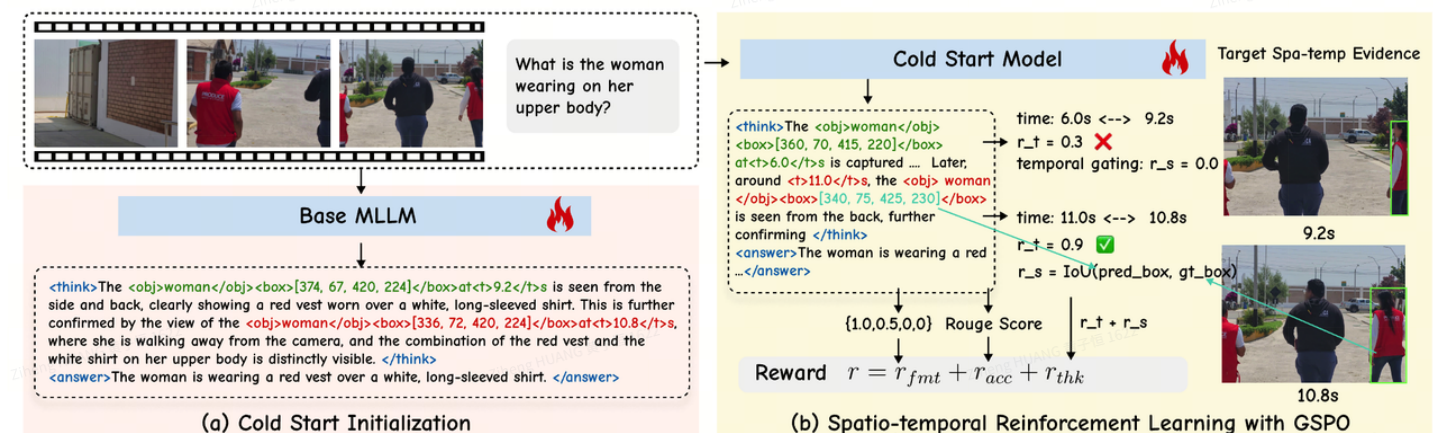
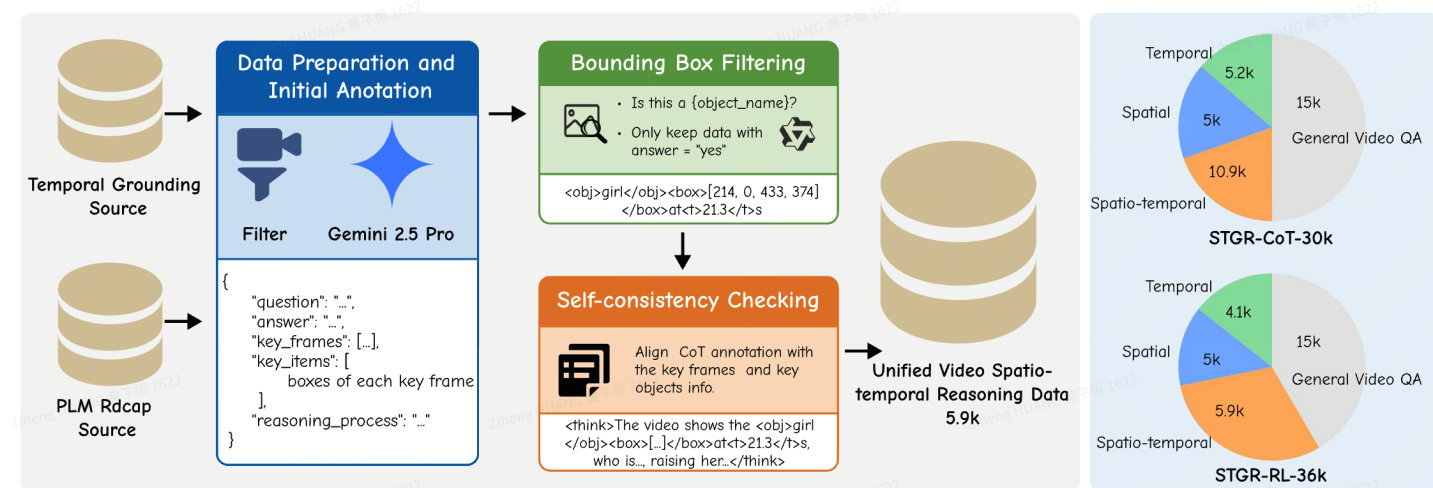
VLM有一个固定的"视觉带宽"——它能处理的视觉token总数是有上限的（context length约束）问题是：

一张4K图像 (3840×2160)—— ViT编码 (patch_size=14)——理论上需要: $(3840/14) \times (2160/14) \approx 42,000$ 个token

实际能处理: ~1,000 个token (context length约束)——实际降采样输入: 约 $384 \times 384 \rightarrow \sim 750$ 个token

1、Open-o3-Video: Grounded Video Reasoning with Explicit Spatio-Temporal Evidence

开源: <https://marinero4972.github.io/projects/Open-o3-Video/>



$$r_t(x, y) = \begin{cases} \frac{1}{M} \sum_{m=1}^M \mathbf{1}[s^{\text{gt}} \leq t_m \leq e^{\text{gt}}], & \tau_t = \text{Int}, \\ \frac{1}{M} \sum_{m=1}^M \exp\left(-\frac{\Delta t_m^2}{2\sigma^2}\right), & \tau_t = \text{Pt}, \\ 0, & \tau_t = \emptyset. \end{cases}$$

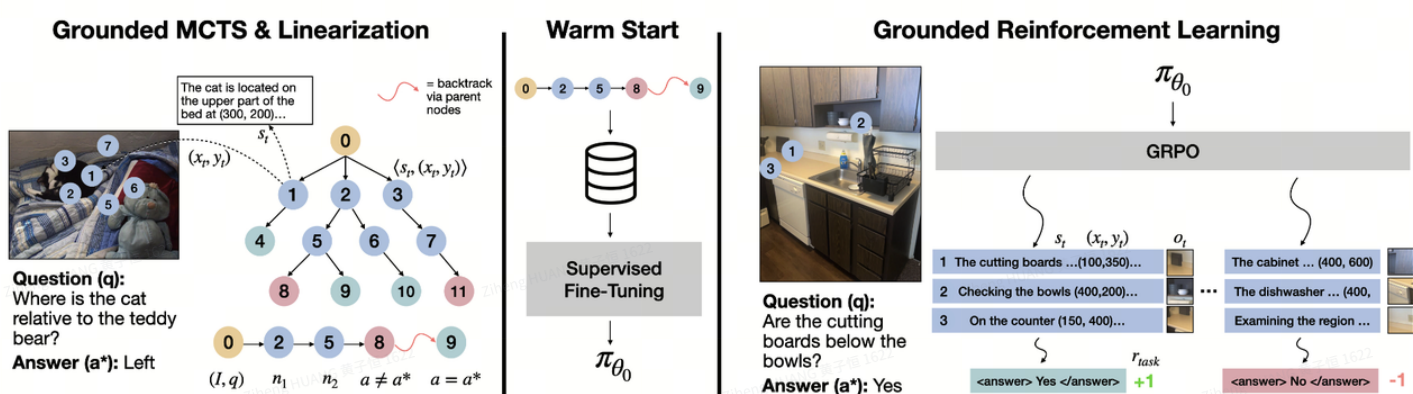
adaptive temporal proximity: 一开始就把时间门槛设得很严，早期训练几乎拿不到奖励；但如果一直设得很松，时间奖励又会很快饱和，模型不会继续把时间戳往真值逼近。所以他们让时间容忍度前期大、后期逐步收紧，先学粗定位，再学精定位

temporal gating: 空间奖励只有在“预测时间已经足够接近真值”时才计算，避免模型在错误时间点上框出某个“看起来显眼但其实不相关”的东西，却仍然拿到空间奖励。

2、Grounded Reinforcement Learning for Visual Reasoning

开源: <https://visually-grounded-rl.github.io/>

Question: The first image is from the beginning of the video and the second image is from the end. How did the camera likely move when shooting the video?	Standard CoT	Vanilla GRPO	Visually Grounded RL
	<think> The first image shows a room with a table, chairs, and televisions mounted on the wall. The second image shows the same room but with different objects and their positions. This suggests that the camera has moved and changed its perspective. </think> <answer> rotated right </answer>	<think> Given that the chair and the bed in Image 1 are in different positions compared to Image 2, the camera must have moved to the right, pointing at Image 1 from Image 2. This is more likely because the chair's new position suggests a visible shift in the scene when the video starts. </think> <answer> rotated right </answer>	<think> To determine how the camera moved, let's analyze the objects' positions relative to each other. The chair in the first image is near the left edge (40, 350), while in the second image, it's more centrally located (700, 400) ... The table in the first image is more towards the left side (200, 250), whereas in the second image, it's shifted towards the center (800, 300). ... Considering the position of the door at (600, 400) in the first image, it is positioned towards the right side of the frame. In the second image, the door is no longer visible ... this aligns with a rotation to the left. </think> <answer> rotated left </answer>



树里的每个节点对应一个 reasoning step (<think> 或 <answer>)，根节点是原始问题；节点会维护当前文本、已经访问过的坐标、价值估计和访问次数。搜索过程还是标准四步：Selection、Expansion、Rollout、Backpropagation。Expansion 时每个节点最多生成 3 个子节点；Rollout 用 VLM 自己继续推理到终局，再由 judge 按最终答案给 0/1 奖励；最后把这个奖励往树上回传

Table 1: Performance (mean $\pm$ 95% CI) on ScreenSpot and VisualWebArena. Multi-t = multi-turn RL with visual feedback.

Model	ScreenSpot -V2	ScreenSpot -Pro	VWA (Vision Only)
<i>Proprietary Models</i>			
GPT-4o	18.1	0.8	19.8 (w/ text)
Claude Comp. Use	-	17.1	-
<i>Open-source Models</i>			
Qwen2-VL-7B	-	1.6	2.9 (w/ text)
Kimi-VL-16B-MoE	92.8	34.5	-
<i>Web Grounding Models</i>			
SeeClick	55.1	1.1	-
CogAgent-18B	-	7.7	-
OS-Atlas-4B	71.9	3.7	-
OS-Atlas-7B	84.1	18.9	-
ShowUI-2B	-	7.7	-
UGround-7B	-	16.5	-
UGround-V1-7B	-	31.1	-
UI-TARS-2B	84.7	27.7	-
UI-R1-3B	-	17.8	-
ICAL-7B	-	-	8.2 (w/ text)
<i>Method Comparison</i>			
Qwen2.5-VL-3B	68.4 (± 2.6)	23.9	4.2 (± 1.3)
+ SFT direct	80.6 (± 2.2)	25.0 (± 2.1)	4.5 (± 1.4)
+ Vanilla GRPO	84.4 (± 2.0)	29.0 (± 2.2)	4.8 (± 1.4)
ViGoRL-3b (Ours)	86.5 (± 1.9)	31.1 (± 2.3)	6.4 (± 1.5)
Multi-t ViGoRL-3b (Ours)	86.1 (± 1.9)	32.3 (± 2.4)	-
Qwen2.5-VL-7B	73.6 (± 2.4)	29.0	5.5 (± 1.5)
ViGoRL-7b (Ours)	91.0 (± 1.6)	33.1 (± 2.3)	11.2 (± 2.1)

Table 2: Accuracy (mean $\pm$ 95% CI) for spatial reasoning.

Model	SAT-2 Val	BLINK	RoboSpatial
<i>Proprietary Models</i>			
GPT-4o	-	60.0	76.2
GPT-4 Turbo	-	54.6	-
Claude 3 Opus	-	44.1	-
Gemini Pro 1.0	-	45.2	-
<i>General Open-source Models</i>			
LLaVA v1.6 34B	-	46.8	-
instructBLIP 13B	-	42.2	-
Molmo	-	-	67.1
<i>Method Comparison</i>			
Qwen2.5VL-3B	46.1 (± 1.5)	44.4 (± 2.3)	54.4 (± 6.5)
+ SFT direct	58.3 (± 1.5)	46.4 (± 2.3)	62.3 (± 6.3)
+ Vanilla GRPO	50.0 (± 1.6)	46.5 (± 2.3)	69.7 (± 6.0)
ViGoRL-3b (Ours)	62.9 (± 1.5)	48.5 (± 2.3)	67.1 (± 6.1)
Qwen2.5-VL-7B	52.6 (± 1.6)	52.9 (± 2.3)	46.7 (± 9.5)
ViGoRL-7b (Ours)	67.5 (± 1.5)	54.1 (± 2.3)	76.4 (± 7.5)

概述

一、图片 Zoom-In 方案

1.1 Training-Free（无需训练）

方法	时间	会议	核心思路	Zoom 实
V* (SEAL)	Dec 2023	CVPR 2024	LLM引导的视觉搜索	递归四分 优先级逐
ZoomEye	Nov 2024	EMNLP 2025 Oral	树搜索模拟人类zoom	图像建树 zoom子图 GPT-4o
MLLMs Know Where to Look	Feb 2025	ICLR 2025	利用MLLM内部注意力定位	基于atten crop再输
DyFo	Apr 2025	CVPR 2025	MCTS动态聚焦	Monte Ca zoom/pa
Zoom-Refine	Jun 2025	—	MLLM自定位+自精炼	模型预测 推理，无
HiDe	Oct 2025	—	重新思考zoom-in（背景干扰才是关键）	Token注 持解耦，
Q-Zoom	Apr 2026	—	查询感知自适应感知	动态门控 1024 tok

1.2 RL训练（强化学习）

方法	时间	核心思路	Zoom 实现
DeepEyes	May 2025	RL端到端学习interleaved多模态CoT	推理中调用zoor bbox→crop回推
Chain-of-Focus	May 2025	SFT+RL自适应zoom决策	多轮生成：输出 新回答
VLM-R3	May 2025	区域识别+推理+精炼	R-GRPO奖励选优 zoom交织CoT
ViGoRL	May 2025	视觉锚定RL	每步发射坐标→
ACTIVE-O3	May 2025	纯RL主动感知	学习sensing策略
CropVLM	Nov 2025	轻量外接crop模块(256M)	RL训练，即插即
AdaptVision	Dec 2025	粗到精自适应	先1/4分辨率，再 crop
TikArt	Feb 2026	Think-Aperture-Observe循环	Zoom(矩形crop Segment(SAM2

二、视频 Zoom-In 方案

2.1 时间维度 Zoom（Temporal Zoom）

方法	时间	会议	核心思路	Zoo
VideoTree	May 2024	CVPR 2025	查询自适应层次树	相关
FOCUS	Oct 2025	—	多臂赌博机关键帧选择	先定帧，
VideoZoomer	Dec 2025	ICLR 2026投稿	RL学习的时间zoom	粗粒→高
Zoom-Zero	Dec 2025	—	粗到精时间grounding	定位帧，Crec
VideoExplorer	Jun 2025	—	"Think with Videos"迭代感知	子问展感

2.2 空间+时间 Zoom（Spatial-Temporal Zoom）

方法	时间	会议	核心思路	Zo
LOVE-R1	Sep 2025	—	自适应空间-时间zoom+多步推理	低:zo
VideoMind	Mar 2025	ICLR 2026	Chain-of-LoRA四角色agent	Gr 辨 GP
Open-o3-Video	Oct 2025	—	显式时空证据推理	冷/间/
VideoChat-R1.5	Sep 2025	NeurIPS 2025	视觉测试时缩放(VTTS)	迭 时